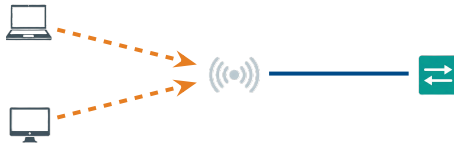


Wireless Networking Fundamentals

Module 12.0 – Deep Dive

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Intro to Networking



Review: From Wired to Wireless Physical Layers

- **Layer 1 & 2 Shift:** We are moving from IEEE 802.3 (Ethernet) to IEEE 802.11 (WLAN).
- **Guided vs. Unguided Media:** Instead of voltages on copper or photons in glass fiber (guided), data is modulated onto electromagnetic waves broadcasted into open space (unguided).
- **Collision Domains:** A wired switch port is a collision domain of one (Full-Duplex). A wireless channel is a massive, shared collision domain (Half-Duplex) acting logically like a legacy Ethernet hub.

The Core Challenge

In wired networks, the cable protects the signal. In wireless, the medium is hostile, shared, and constantly subject to physical interference and atmospheric degradation.

Learning Outcomes (Network+ Objectives)

Mapping directly to CompTIA Network+ domains, you will be able to:

- **Analyze** fundamental RF characteristics: frequency, amplitude, and phase modulation.
- **Calculate** the impact of overlapping vs. non-overlapping channels in 2.4GHz, 5GHz, and 6GHz bands.
- **Deconstruct** the CSMA/CA protocol, including the RTS/CTS mechanism for resolving the hidden node problem.
- **Differentiate** 802.11 iterations based on modulation techniques (DSSS, OFDM, OFDMA) and spatial streaming (MIMO).
- **Design** physical AP layouts utilizing appropriate antenna polarizations and radiation patterns (Omni vs. Yagi/Patch).

Section 1

Basic Wireless Concepts

The Wireless Medium: Radio Frequency (RF)

- **Carrier Waves:** Access Points generate a continuous sine wave (the carrier).
- **Modulation:** Data (1s and 0s) is superimposed onto the carrier wave by rapidly altering its state.
- **QAM (Quadrature Amplitude Modulation):** Modern Wi-Fi alters *both* the phase and the amplitude simultaneously to encode multiple bits per symbol (e.g., 256-QAM or 1024-QAM).
- **Environmental Variables:** Unlike a Cat6 cable, an RF signal's SNR (Signal-to-Noise Ratio) constantly fluctuates based on moving physical objects and competing transmitters.

The Half-Duplex Reality

A wireless radio uses the exact same antenna to transmit and receive. It cannot do both at once without deafening itself.

Therefore, **all Wi-Fi is strictly half-duplex.**

RF Characteristics: Frequency, Wavelength, and Amplitude

- **Frequency (f):** The number of full wave cycles per second, measured in Hertz (Hz). Wi-Fi operates in the Gigahertz (GHz) range (billions of cycles per second).
- **Wavelength (λ):** The physical distance between two consecutive peaks of a wave.

The Speed of Light Equation

$$c = \lambda f$$

Because the speed of light (c) is constant, frequency and wavelength are inversely related. As frequency increases, wavelength gets shorter.

- **Amplitude:** The power/strength of the wave, often measured in decibel-milliwatts (dBm).
- Lower frequency (longer wavelength) waves penetrate solid objects much better than high-frequency waves.

The 2.4 GHz vs. 5 GHz Bands

Metric	2.4 GHz ISM Band	5 GHz UNII Bands
Wavelength	≈ 4.9 inches (Longer)	≈ 2.3 inches (Shorter)
Attenuation	Excellent at penetrating drywall, brick, and wood.	Absorbed quickly by walls; heavily attenuated by water.
Capacity	Only 3 non-overlapping channels (1, 6, 11).	Up to 24 non-overlapping 20MHz channels.
Interference	Highly congested (Microwaves, Bluetooth, IoT, neighbors).	Relatively clean, though some channels require DFS (radar avoidance).
Max Throughput	Low (Usually constrained to 20MHz widths).	High (Supports 40MHz, 80MHz, and 160MHz channel bonding).

The New Frontier: 6 GHz and Wi-Fi 6E/7

- **The Spectrum Starvation Problem:** Enterprise 5 GHz networks ran out of space when trying to bond channels for higher gigabit speeds.
- **The 6 GHz Solution:** The FCC opened 1,200 MHz of entirely new, contiguous spectrum for Wi-Fi 6E and Wi-Fi 7.
- Yields **59** new 20 MHz channels, or 7 massive 160 MHz channels.

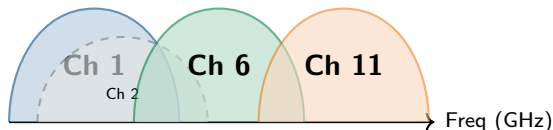
Greenfield Deployment

Unlike 2.4/5 GHz, the 6 GHz band strictly prohibits legacy standards (802.11a/b/g/n/ac). Only highly efficient 802.11ax/be devices are allowed.

Security Constraint: WPA3 is strictly mandatory on the 6 GHz band. You cannot configure an open or WPA2 6 GHz SSID.

Channels and Co-Channel Interference

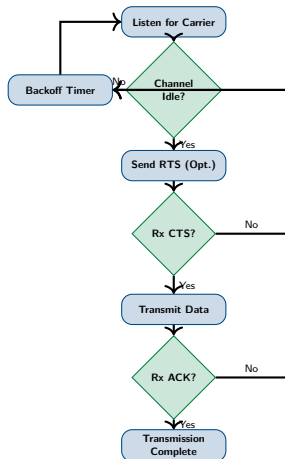
- The 2.4GHz band is 72 MHz wide, divided into 11 channels (in the US).
- **The Math Problem:** Each 802.11 standard channel requires roughly 20-22 MHz of frequency space, but the channels are drawn only **5 MHz apart**.
- Therefore, Channel 2 physically overlaps with 1, 3, 4, and 5.
- **CCI (Co-Channel):** APs on the exact same channel politely wait their turn.
- **ACI (Adjacent Channel):** APs on overlapping channels (e.g., 1 and 2) talk over each other as unintelligible RF noise, causing massive packet loss.



2.4 GHz rule:
Use channels **1, 6, and 11** only.

CSMA/CA & The Hidden Node Problem

- **CSMA/CA** relies on a Clear Channel Assessment (CCA).
- **Hidden Node Problem:** Laptop A and Laptop B can both reach the AP, but are too far apart to hear *each other*. They might transmit simultaneously, causing a collision at the AP.
- **Solution (RTS/CTS):** Instead of just sending data, the device sends a tiny Request-To-Send. The AP replies with a Clear-To-Send, which tells *all* other devices to set their Network Allocation Vector (NAV) timer and be quiet.



Section 2

Wireless Standards (802.11)

The Evolution of Wi-Fi: Standards and Generations

- The **IEEE 802.11** working group defines the PHY (Physical) and MAC (Media Access Control) layers for wireless LANs.
- The **Wi-Fi Alliance** introduced generational names (Wi-Fi 4, 5, 6) to simplify marketing and consumer understanding.

IEEE Standard	Wi-Fi Name	Frequency Band(s)	Max Theo. Speed	Key Physical Innovation
802.11b (1999)	Wi-Fi 1	2.4 GHz	11 Mbps	DSSS (Direct Sequence Spread Spectrum)
802.11a (1999)	Wi-Fi 2	5 GHz	54 Mbps	OFDM (Orthogonal Frequency Division)
802.11g (2003)	Wi-Fi 3	2.4 GHz	54 Mbps	Brought OFDM to the 2.4 GHz band
802.11n (2009)	Wi-Fi 4	2.4 GHz & 5 GHz	600 Mbps	MIMO (Multiple Input, Multiple Output)
802.11ac (2013)	Wi-Fi 5	5 GHz only	6.9 Gbps	MU-MIMO (Downlink), 160MHz Channels
802.11ax (2019)	Wi-Fi 6 / 6E	2.4, 5, & 6 GHz	9.6 Gbps	OFDMA (Resource Units), BSS Coloring
802.11be (2024+)	Wi-Fi 7	2.4, 5, & 6 GHz	46 Gbps	MLO (Multi-Link Operation), 320MHz Ch.

Legacy Standards: 802.11a, b, and g

The Pioneers

- **802.11b:** Used **DSSS** (Direct Sequence Spread Spectrum). Highly resilient to interference but very slow (11 Mbps). Ignited the Wi-Fi revolution.
- **802.11a:** Operated exclusively in the 5 GHz band. Used **OFDM**, which splits the signal across multiple subcarriers. Much faster (54 Mbps) but lacked range.
- **802.11g:** The grand compromise. Brought the speed of OFDM (54 Mbps) down to the 2.4 GHz band. Backwards compatible with 802.11b devices.

The Problem with Legacy

Because 802.11g had to remain backwards compatible with 802.11b, it introduced **protection mechanisms** (like sending RTS/CTS frames at 1 Mbps) so older devices wouldn't accidentally transmit.

This overhead destroyed network efficiency.

Best Practice: Disable legacy 802.11b rates (1, 2, 5.5, 11 Mbps) on modern enterprise networks.

802.11n (Wi-Fi 4): Introducing MIMO

- **802.11n** revolutionized Wi-Fi by introducing **MIMO (Multiple Input, Multiple Output)**.
- Previous standards suffered from **Multipath Interference** (signals bouncing off walls and arriving out of phase, destroying the data).
- MIMO uses multiple antennas and complex digital signal processing to treat multipath echoes as a *benefit*, reconstructing the separated streams.
- **Spatial Multiplexing:** Sending multiple independent data streams simultaneously over the exact same frequency channel.

Antenna Configurations

You will see APs listed as **2x2:2** or **4x4:4**.

- First number: Transmit antennas (Tx)
- Second number: Receive antennas (Rx)
- Third number: Spatial Streams

A 3x3:3 AP can send three distinct streams of data to a 3x3:3 laptop, effectively tripling the throughput compared to legacy standards.

802.11ac (Wi-Fi 5): Wider Channels and MU-MIMO

- **5 GHz Exclusive:** 802.11ac abandoned the 2.4 GHz band to focus entirely on the wider, cleaner 5 GHz spectrum.
- **Channel Bonding:** Allowed bonding 20 MHz channels into 40, 80, or even 160 MHz contiguous blocks, acting like a wider highway for massive throughput.
- **Downlink MU-MIMO:** (Multi-User MIMO). While 802.11n could only talk to one client at a time (SU-MIMO), 802.11ac APs could form separate RF beams to talk to *multiple different clients* at the exact same millisecond.

The MU-MIMO Limitation

In 802.11ac, MU-MIMO only worked on the **downlink** (AP to clients). Furthermore, clients had to be physically separated so the AP's mathematical "nulls" (RF blind spots) could keep the streams from interfering.

QAM Upgrade

Moved from 64-QAM to **256-QAM**, cramming more bits into every RF symbol, provided the client was very close to the AP (high SNR).

Modern Standards: 802.11ax (Wi-Fi 6) & 802.11be (Wi-Fi 7)

802.11ax (Wi-Fi 6 & 6E)

- **Goal: Efficiency.** Older Wi-Fi was fast for one user; Wi-Fi 6 is fast for *hundreds* of users simultaneously.
- **OFDMA:** Chops a standard 20 MHz channel into smaller **Resource Units (RUs)**. The AP can now pack payloads for a laptop, a phone, and a smart thermostat into a single transmission.
- **BSS Coloring:** Adds a tag to frames so APs on the same channel can ignore each other's traffic, reducing Co-Channel Interference delays.

802.11be (Wi-Fi 7)

- **Goal: Extreme Throughput.** Designed to replace wired Ethernet for VR and uncompressed video.
- **MLO (Multi-Link Operation):** A Wi-Fi 7 device can connect to the 5 GHz and 6 GHz bands *simultaneously*, aggregating the bandwidth and dropping latency to near-zero.
- **4096-QAM:** Packs 12 bits per symbol (requires pristine RF conditions).

Case Study: Dustin's Airtime Anomaly

► Case Study: The Legacy Device Drag

Dustin Henderson, Senior RF Engineer at Hawkins National Lab, is frustrated. The lab just spent \$50,000 upgrading the research wing to state-of-the-art **Wi-Fi 6 (802.11ax)** access points.

However, researchers are complaining that the network periodically crawls to a halt. Dustin runs a packet capture and discovers that a single, ancient barcode scanner operating on **802.11b (11 Mbps)** is actively connecting to the 2.4 GHz network to inventory test tubes.

Review Questions

- ❶ What frequency band is the barcode scanner using?
- ❷ Why would a single slow device cause the brand-new Wi-Fi 6 laptops to slow down?
- ❸ What configuration change should Dustin make to the APs to permanently resolve this?

Case Study Solution: Dustin's Airtime Anomaly

✓ Solution: The Legacy Device Drag

- ① The 802.11b scanner operates exclusively on the **2.4 GHz band**.
- ② **Airtime Fairness / The Slow Client Problem:** Wi-Fi is a shared, half-duplex medium. It takes the 802.11b scanner a massive amount of "airtime" to send a tiny payload (because it is transmitting at 1 or 11 Mbps). While the slow scanner is occupying the channel, the Wi-Fi 6 laptops must wait in the CSMA/CA queue.
- ③ **Disable Legacy Data Rates:** Dustin must configure the Wireless LAN Controller to disable the 802.11b mandatory data rates (1, 2, 5.5, and 11 Mbps).

Key Takeaway

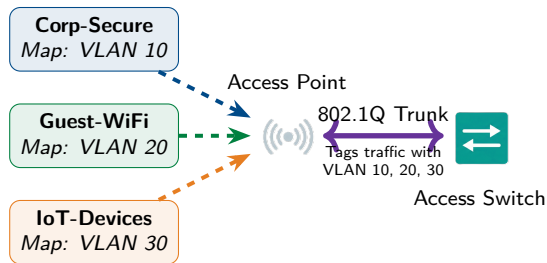
Disabling legacy data rates ensures the AP will simply refuse to associate with 802.11b devices, protecting the airtime for modern, highly efficient clients. Hawkins Lab will have to buy a new barcode scanner.

Section 3

Enterprise Wireless Design

Logical Design: SSIDs and VLAN Mapping

- An **SSID (Service Set Identifier)** is the network name broadcasted by the AP.
- In an enterprise, an AP typically broadcasts multiple SSIDs simultaneously (e.g., Corp-Secure, Guest-WiFi, IoT-Devices).
- Logically, each SSID is mapped to a specific **VLAN** on the wired network.
- The connection between the AP and the access switch must be an **802.1Q Trunk** to carry the separated VLAN traffic.



Autonomous vs. Lightweight Architecture

Autonomous APs (Fat APs)

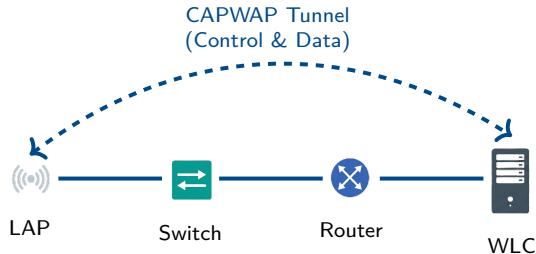
- Standalone, self-contained devices (like a standard home router).
- Each AP contains its own management, security protocols, routing intelligence, and QoS rules.
- **The Scaling Problem:** If an enterprise has 500 APs, a password change or firmware update requires logging into 500 separate management interfaces. *It does not scale.*

Lightweight APs (Thin APs)

- Stripped-down radios that do little more than broadcast RF and forward frames.
- All "brains" (authentication, roaming decisions, channel assignments, security) are moved to a centralized **Wireless LAN Controller (WLC)**.
- **The Solution:** You configure the WLC once, and it pushes the configuration dynamically to all 500 APs instantly.

The WLC and CAPWAP Tunnels

- Lightweight APs communicate with the WLC using **CAPWAP** (Control and Provisioning of Wireless Access Points), UDP ports 5246/5247.
- **Split-MAC Architecture:**
 - **AP handles real-time tasks:** Beacons, ACKs, and RF transmission.
 - **WLC handles management:** Authentication, roaming, and security.
- **Data Tunneling:** In most enterprise designs, all user Wi-Fi traffic is tunneled directly to the WLC before being dropped onto the wired LAN.



Design Note

Because all traffic flows through the WLC, the WLC must be placed at a high-bandwidth choke point (like the Core or Data Center).

Physical Design: The Site Survey

- A **Site Survey** is the critical process of planning AP placement to ensure adequate coverage, capacity, and minimal interference.

Predictive Survey

Performed off-site using CAD floorplans and RF simulation software (e.g., Ekahau). Walls are drawn and assigned attenuation values (brick vs. drywall) to mathematically estimate AP coverage.

Active Survey

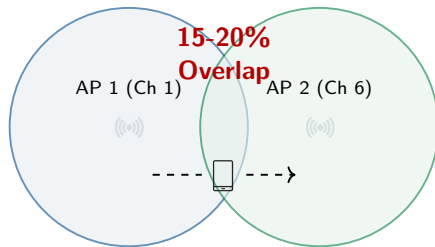
An engineer walks the actual building with a testing laptop, actively connecting to test APs ("AP on a stick") to measure real-world throughput, latency, and packet loss.

Passive Survey

Walking the building with a spectrum analyzer purely to listen. Used to map existing RF noise, detect rogue APs, and identify non-Wi-Fi interference (microwaves, Bluetooth).

Antenna Types and Coverage Cells

- **Omnidirectional (Dipole):** Radiates RF energy equally in a 360-degree donut shape. Ideal for ceiling mounts in the center of standard rooms.
- **Directional (Patch/Yagi):** Focuses RF energy in a specific direction, increasing gain (reach) in that direction while reducing it elsewhere. Ideal for long hallways, high warehouse ceilings, or bridging buildings.



Note: Adjacent APs must operate on non-overlapping channels. For 2.4 GHz, use 1 and 6 to prevent ACI in the roaming zone.

Seamless Roaming

As a user walks, their device must seamlessly **roam** from AP 1 to AP 2. This requires exactly **15% to 20% coverage overlap** between the RF cells.

Case Study: Robin's Starcourt Site Survey

► Case Study: The Food Court Dilemma

Robin Buckley is designing the AP layout for Starcourt Mall. She needs dense coverage for the food court and must keep the back-office signal from bleeding into the parking lot.

Review Questions

- 1 For the dense food court, should Robin use a few high-powered APs or many low-powered APs? Why?
- 2 What specific type of antenna should Robin deploy near the glass doors to stop the signal from bleeding into the parking lot?

Case Study Solution: Robin's Starcourt Site Survey

✓ Solution: The Food Court Dilemma

- ➊ **Food Court Density:** She should deploy **many low-powered APs**. High-powered APs create massive coverage cells, forcing hundreds of devices to fight for airtime on a single channel (a CSMA/CA nightmare). Small, low-power cells (Microcells) allow Robin to reuse channels, dramatically increasing total bandwidth capacity.
- ➋ **Parking Lot Bleed:** Robin should use **Directional (Patch) antennas** mounted on the interior perimeter walls, aimed *inward* toward the mall. Standard omnidirectional antennas would radiate 360 degrees, sending half their RF energy straight through the glass into the parking lot.

Key Takeaway

Good wireless design isn't about making the signal as loud as possible; it is about strictly controlling the shape, size, and boundaries of the RF cell.

Section 4

Wireless Security & Access

The Evolution of Wireless Encryption

- RF data is broadcasted in open air. Without encryption, anyone with a protocol analyzer (like Wireshark) can capture and read all network traffic.

Protocol	Cipher / Encryption	Key Vulnerability	Modern Status
WEP	RC4 (Weak 24-bit IVs)	IVs repeat; can be cracked in minutes by capturing traffic.	Dead. Never use under any circumstance.
WPA	TKIP (RC4 wrapper)	Band-aid fix designed to run on old WEP hardware.	Deprecated. Easily compromised.
WPA2	AES / CCMP	Dictionary/Brute-force attacks against the Pre-Shared Key (4-way handshake).	Current Standard. Secure if the password is long/complex.
WPA3	AES / GCMP	None widely known yet. Uses SAE instead of PSK.	The Future. Mandatory for Wi-Fi 6E (6 GHz).

WPA3 and Forward Secrecy

WPA3's SAE (Simultaneous Authentication of Equals) provides **Forward Secrecy**. If an attacker records encrypted Wi-Fi traffic today, and steals the password tomorrow, they *still* cannot decrypt the old traffic.

Authentication: Personal vs. Enterprise

WPA2 / WPA3 - Personal (PSK)

- Uses a **Pre-Shared Key** (password).
- Every single device on the network types in the exact same password.
- **The Fatal Flaw for Business:** If an employee is fired, you must change the Wi-Fi password on *every single device* in the company to properly revoke their access.
- Suitable only for homes and small coffee shops.

WPA2 / WPA3 - Enterprise (802.1X)

- Uses **802.1X** and a back-end **RADIUS** server (tied to Active Directory).
- Users authenticate with their own unique username/password, or a digital certificate.
- **The Enterprise Benefit:** If someone is fired, you disable their specific AD account. Their Wi-Fi access is instantly revoked, and nobody else is affected.

Managing Guest Access and BYOD

- **BYOD (Bring Your Own Device)** policies allow employees to use personal smartphones. Because IT does not control these devices, they must be treated as untrusted (placed on an isolated VLAN).
- **MDM (Mobile Device Management):** Software profiles installed on personal devices that enforce security policies (PIN required, encrypted storage) before granting access to corporate Wi-Fi and email.

Captive Portals

- A web page that intercepts HTTP/DNS traffic until a user completes an action.
- Used to enforce an **Acceptable Use Policy (AUP)**, process payments, or verify an email address.
- **Warning:** Networks with Captive Portals are typically **Open** (no Layer 2 encryption). Traffic can be intercepted easily.

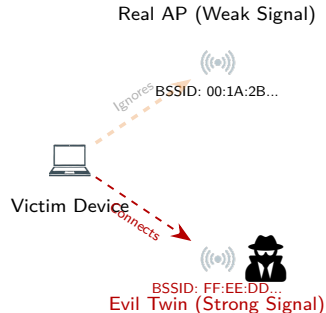
Wireless Threats: Rogue APs and Evil Twins

Rogue Access Point

- Unauthorized AP plugged into the corporate network.
- Often installed by an employee for convenience.
- **Danger:** Bypasses firewall, RADIUS, and WLC.

Evil Twin Attack

- Malicious AP spoofing the legitimate corporate SSID.
- Devices roam to the strongest signal, letting the attacker intercept traffic.



Defense

Use a **WIDS/WIPS** to scan for spoofed MAC addresses and unauthorized APs.

Case Study: Steve's Evil Twin

► Case Study: The Stolen Credentials

Steve Harrington sees staff phones connect to an open guest network named "Starcourt_Staff" with a captive portal. A second AP with the same SSID is sitting in a van outside the breakroom and blasting a much stronger signal.

Review Questions

- 1 What specific type of attack is the van in the parking lot executing?
- 2 Why are the employee's personal phones automatically connecting to the van's AP instead of the real one inside the mall?
- 3 How would moving to **WPA2-Enterprise (802.1X)** instead of an Open Captive Portal prevent this attack?

Case Study Solution: Steve's Evil Twin

✓ Solution: The Stolen Credentials

- ➊ **Evil Twin:** The attacker in the van is spoofing the legitimate network name and serving a fake Captive Portal page to harvest credentials.
- ➋ **Signal Strength / Roaming:** Wireless client drivers are programmed to aggressively roam to the AP with the strongest RF signal. Because the van is pumping out high power near the breakroom, the phones switch to it automatically.
- ➌ **802.1X Mutual Authentication:** If the network used 802.1X (EAP-TLS or PEAP), the authentication process requires mutual trust. The employee's phone would challenge the van's AP to present a valid digital certificate signed by the corporate CA. Because the van doesn't have it, the phone silently aborts the connection, keeping the credentials safe.

Key Takeaway

Open networks with Captive Portals are trivial to spoof and provide no Layer 2 encryption. 802.1X secures BYOD access by cryptographically verifying both the user *and* the network infrastructure.

Section 5

Wireless Troubleshooting

Environmental Issues and RF Attenuation

- When troubleshooting wireless coverage gaps, the physical environment is the primary suspect.
- **Attenuation:** The natural loss of signal strength as an RF wave travels through space or passes through materials.

RF Behaviors

- **Absorption:** Materials (concrete, brick, human bodies) absorb RF energy, killing the signal.
- **Reflection:** Metal objects (filing cabinets, elevator shafts) bounce the signal, causing multipath interference.
- **Refraction:** Glass and water can bend the RF wave, changing its trajectory.

Non-Wi-Fi Interference

- Wi-Fi runs in unlicensed ISM bands.
- Baby monitors, microwaves, cordless phones, and Bluetooth all share the 2.4 GHz band.
- Because CSMA/CA requires devices to wait for "clear air", non-Wi-Fi interference doesn't just corrupt data—it freezes the channel completely.

The Rule of 3s and 10s (dB)

In RF math:

- +3 dB doubles the power.
- -3 dB halves the power.
- +10 dB increases power by 10x.

Troubleshooting Metrics: RSSI and SNR

- **RSSI (Received Signal Strength Indicator):** A measurement of how well your device can hear the AP. Measured in **negative dBm**.
- Closer to zero is better.
- **Noise Floor:** The background ambient RF noise (microwaves, cosmic radiation, neighbor's Wi-Fi).
- **SNR (Signal-to-Noise Ratio):** The difference between the RSSI and the Noise Floor. A high SNR means the signal is clearly distinguishable from the noise.

RSSI Value	Expected Performance
-30 dBm	Perfect (Right next to AP)
-65 dBm	Excellent (Enterprise standard)
-70 dBm	Good (Reliable packet delivery)
-80 dBm	Poor (Dropped packets, slow)
-90 dBm	Unusable (Disconnects)

Calculating SNR

Signal: -65 dBm

Noise Floor: -90 dBm

SNR = 25 dB (*Excellent*)

*If SNR drops below 10 dB, the device cannot decode the transmission.

Common Configuration and Placement Issues

Improper AP Placement

- **Too High:** Mounting APs on 30-foot warehouse ceilings puts clients too far away.
- **Hidden APs:** Placing APs above drop ceilings near metal HVAC ducts severely attenuates and reflects the signal.
- **Wall Mounts:** Mounting omnidirectional APs sideways on a wall instead of flat on a ceiling pushes half the signal into the floor/ceiling.

Logical Configuration Errors

- **Channel Overlap:** Manually assigning an AP to Channel 3 on the 2.4 GHz band causes crippling Adjacent Channel Interference.
- **Capacity Overload:** Connecting 100 students to a single AP in an auditorium. Wi-Fi is shared; airtime becomes exhausted.
- **Frequency Mismatch:** A legacy 2.4 GHz IoT device cannot see an SSID that is only broadcasting on 5 GHz.

Case Study: Hopper's Microwave Mystery

► Case Study: The Lunchtime Latency

Jim Hopper, IT Director at the Hawkins Police Station, is aggressively troubleshooting a massive latency spike that happens every day exactly at 12:30 PM on the dispatch floor.

During this window, wireless IP phones drop calls, and the dispatch software running on 2.4 GHz laptops disconnects entirely. Checking the WLC logs, Hopper sees that the AP serving the dispatch floor is experiencing 90% channel utilization and massive packet retries, despite only having 5 connected devices.

Review Questions

- 1 What phenomenon is occurring every day at 12:30 PM?
- 2 Why does this non-Wi-Fi interference cause Wi-Fi packet retries and high channel utilization?
- 3 What are two potential solutions Hopper can implement to fix the dispatch network?

Case Study Solution: Hopper's Microwave Mystery

✓ Solution: The Lunchtime Latency

- ❶ **Non-Wi-Fi Interference:** Joyce is heating up her lunch. Microwave ovens operate at 2.4 GHz. A leaky, unshielded microwave acts as a massive RF jammer on the 2.4 GHz band.
- ❷ **CSMA/CA Breakdown:** Wi-Fi devices must perform a Clear Channel Assessment (CCA). When the microwave is running, the channel is never clear. Devices wait indefinitely (latency). When they do try to transmit, the RF noise corrupts the frame, no ACK is received, and the packet must be retried.
- ❸ **Solutions:**
 - *Solution A:* Move the dispatch laptops and IP phones to the **5 GHz band**, completely avoiding the microwave's frequency.
 - *Solution B:* Throw the leaky microwave away and buy a properly shielded one for the breakroom.

Module 12.0 Summary

Wireless Fundamentals

- RF uses 2.4 GHz (range/crowded), 5 GHz (speed/clean), and 6 GHz (Wi-Fi 6E/7).
- Wi-Fi uses CSMA/CA to avoid collisions.
- 802.11 evolved from DSSS (802.11b) to MIMO (802.11n) to OFDMA (802.11ax).

Security & Threats

- WPA3 uses SAE for forward secrecy.
- 802.1X (Enterprise) ties access to identity.
- Rogue APs bypass firewalls; Evil Twins spoof real networks to steal credentials.
- BYOD requires MDM or isolated VLANs.

Enterprise Design

- WLCs manage lightweight APs via CAPWAP tunnels for centralized control.
- Antenna choice (Omni vs. Directional) shapes the RF cell to optimize coverage.

Troubleshooting

- Good RSSI (-65 dBm) and high SNR are critical.
- Non-Wi-Fi interference destroys 2.4 GHz.
- Only use channels 1, 6, and 11 on 2.4 GHz.